

Testing & Quality Assurance Strategy

Source: docs/TESTING.md • Status: Verified & Synchronized

Product: Cubyntra

Organization: Necookie Labs

Status: Implemented & Passing (35/35 Tests)

Directory: tests/

1. Quality Assurance Philosophy

Because Cubyntra operates at the intersection of discrete group theory, computer vision heuristics, and 3D coordinate geometry, automated testing is mandatory for ensuring system stability.

Core Testing Invariants:

- Mathematical Invariant Rigor: Scramble operations must never corrupt piece counts or violate solvability parity unless specifically testing rejection paths.
- Zero Regressions: All core modules (src/cube, src/vision, src/solver) must maintain full unit test coverage.
- Deterministic Integration Lifecycle: Integration tests must simulate full end-to-end user workflows from raw frame sampling to final solved identity state.

2. Test Suite Architecture

Cubyntra utilizes Vitest for ultrafast, ESM-native TypeScript test execution.

```
tests/
% % % cube.test.ts      # Domain models, face rotations, invariant validation (14 tests)
% % % vision.test.ts    # Color space conversions, Delta-E, stability filtering (9 tests)
% % % solver.test.ts    # Two-phase algorithm, move parsing, verification (7 tests)
% % % integration.test.ts # End-to-end scanning, error recovery, and playback (5 tests)
```

3. Test Coverage Matrix

3.1 tests/cube.test.ts

- Initial Solved State: Verifies exactly 6 faces, 9 stickers per face, correct center colors.
- Single Face Rotations: Verifies U, D, L, R, F, B preserve adjacent face boundary order.
- Double Turns & Inverses: Verifies $M \cdot M' = I$ and $M \cdot M'^2 \cdot M = I$.
- Validation Engine:
 - Rejects invalid facelet counts (e.g. 10 whites, 8 yellows).
 - Rejects malformed strings (< 54 or > 54 chars, unexpected tokens).
 - Rejects physical impossible pieces (e.g. White-Yellow edge or Blue-Green corner).
 - Rejects parity errors (unsolvable single edge flip or single corner twist).
- Scramble Generation: Confirms random scrambles maintain piece invariants.

3.2 tests/vision.test.ts

- Color Conversions:
 - RGB to HSV conversion accuracy across primary color bounds.
 - RGB to CIELAB $L^*a^*b^*$ conversion accuracy under D65 standard illuminant.
 - CIELAB Delta-E Distance: Validates perceptual distance matching between reference swatches and noisy camera inputs.
- Grid Sampling: Verifies 9 sample points centered accurately within camera bounding boxes with circular kernel averaging.

- Temporal Stability Filter:
- Verifies stability count increments on static frames.
- Verifies stability resets to 0 when camera detects motion blur or sticker change.
- Locks face capture only after threshold (10 consecutive frames).

3.3 tests/solver.test.ts

- Identity State: Solved cube returns empty move sequence ([]) in < 1 ms.
- Benchmark Positions:
- Superflip: Solved within 20 HTM moves.
- T-Permutation: Solved accurately.
- Random Scrambles: Solved within 20–24 moves.
- Move Parsing: Parses standard notation (R, U', F2) into structured objects.
- Move Inversion: Correctly reverses move sequences:
 $(R \setminus U \setminus R' \setminus U')^{-1} = U \setminus R \setminus U' \setminus R'$
- Simulation Verification: Proves that applying the generated solution to the initial state arrives at identity.

3.4 tests/integration.test.ts

- Full Scan-to-Solve Lifecycle: Simulates sequential scanning of 6 faces, validates cube state, runs Kociemba solver, and verifies final state.
- Error Recovery Workflow: Simulates an optical misclassification (e.g. White misread as Yellow), verifies validator rejection, applies manual user correction, and solves successfully.
- Interactive Scramble & Solve: Applies random moves to digital twin, solves, and navigates step-by-step playback to completion.

4. Test Execution & CI Automation

```
# Execute full test suite
npm run test

# Run tests in continuous watch mode
npm run test:watch

# Generate code coverage report
npm run test:coverage
```

Current Test Execution Results:

```
' tests/cube.test.ts (14 tests) 18ms
' tests/vision.test.ts (9 tests) 12ms
' tests/solver.test.ts (7 tests) 42ms
' tests/integration.test.ts (5 tests) 28ms

Test Files  4 passed (4)
  Tests    35 passed (35)
    Duration 240ms (transform 48ms, setup 0ms, collect 38ms, tests 100ms, environment 0ms,
prepare 22ms)
```